

How to Read 100 Million Blogs (& Classify Deaths Without Physicians)

Gary King
Harvard University

March 6, 2008

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”
- Gary King and Ying Lu. “Verbal Autopsy Methods with Multiple Causes of Death,” forthcoming, *Statistical Science*

- Daniel Hopkins and Gary King. “Extracting Systematic Social Science Meaning from Text”
- Gary King and Ying Lu. “Verbal Autopsy Methods with Multiple Causes of Death,” forthcoming, *Statistical Science*
- Copies at <http://gking.harvard.edu>

Inputs and Target Quantities of Interest

Inputs and Target Quantities of Interest

- Input Data:

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - individual document classifications (spam filters)
 - proportion in each category (proportion email which is spam)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - individual document classifications (spam filters)
 - proportion in each category (proportion email which is spam)
- Estimation

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)
 - **proportion in each category** (proportion email which is spam)
- Estimation
 - *Can* get the 2nd by counting the 1st (turns out not to be necessary!)

Inputs and Target Quantities of Interest

- Input Data:
 - Large set of text documents
 - A set of (mutually exclusive and exhaustive) categories
 - A small set of documents hand-coded into the categories
- Quantities of interest
 - **individual document classifications** (spam filters)
 - **proportion in each category** (proportion email which is spam)
- Estimation
 - *Can* get the 2nd by counting the 1st (turns out not to be necessary!)
 - High classification accuracy $\Rightarrow$ unbiased category proportions

Blogs as a Running Example

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 69–108 million worldwide now.

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 69–108 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 69–108 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran
- “We are living through the largest expansion of expressive capability in the history of the human race”

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 69–108 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran
- “We are living through the largest expansion of expressive capability in the history of the human race”
- Measures **classical** notion of public opinion: active public expressions designed to influence policy and politics (previously: strikes, boycotts, demonstrations, editorials)

Blogs as a Running Example

- Blogs (web logs): web version of a daily diary, with posts listed in reverse chronological order.
- 8% of U.S. Internet users (12 million) have blogs
- Growth: ≈ 0 in 2000; 69–108 million worldwide now.
- A democratic technology: 6 million in China and 700,000 in Iran
- “We are living through the largest expansion of expressive capability in the history of the human race”
- Measures **classical** notion of public opinion: active public expressions designed to influence policy and politics (previously: strikes, boycotts, demonstrations, editorials)
- (Public opinion $\nRightarrow$ surveys)

One specific quantity of interest

One specific quantity of interest

- Affect about President Bush and 2008 candidates

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:
 - Part ordinal, part nominal categorization

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:
 - Part ordinal, part nominal categorization
 - “Sentiment categorization is more difficult than topic classification”

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:

<u>Label</u>	<u>Category</u>
-2	extremely negative
-1	negative
0	neutral
1	positive
2	extremely positive
NA	no opinion expressed
NB	not a blog

- Hard case:

- Part ordinal, part nominal categorization
- “Sentiment categorization is more difficult than topic classification”
- Informal language: “my crunchy gf thinks dubya hid the wmd's, :)!”

One specific quantity of interest

- Affect about President Bush and 2008 candidates

- Specific categories:
- | <u>Label</u> | <u>Category</u> |
|--------------|----------------------|
| -2 | extremely negative |
| -1 | negative |
| 0 | neutral |
| 1 | positive |
| 2 | extremely positive |
| NA | no opinion expressed |
| NB | not a blog |

- Hard case:
 - Part ordinal, part nominal categorization
 - “Sentiment categorization is more difficult than topic classification”
 - Informal language: “**my crunchy gf thinks dubya hid the wmd's, :)**!”
 - Little common internal structure (no inverted pyramid)

The Conversation about John Kerry's Botched Joke

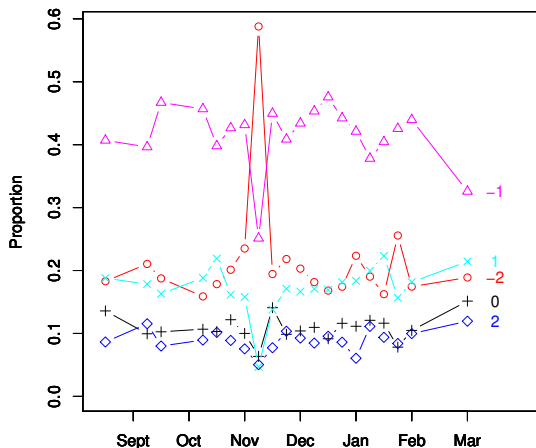
The Conversation about John Kerry's Botched Joke

You know, education — if you make the most of it . . . you can do well. If you don't, you get stuck in Iraq.

The Conversation about John Kerry's Botched Joke

You know, education — if you make the most of it . . . you can do well. If you don't, you get stuck in Iraq.

Affect Towards John Kerry



2006-2007

Representing Text as Numbers

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton:
201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton:
201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.
 - **keep only** unigrams in $> 1\%$ or $< 99\%$ of documents: 3,672 variables

Representing Text as Numbers

- **Filter:** choose English language blogs that mention Bush
- **Preprocess:** convert to lower case, remove punctuation, keep only word stems (“consist”, “consisted”, “consistency” $\rightsquigarrow$ “consist”)
- **Code variables:** presence/absence of unique unigrams, bigrams, trigrams
- **Our Example:**
 - Our 10,771 blog posts about Bush and Clinton: 201,676 unigrams, 2,392,027 bigrams, 5,761,979 trigrams.
 - **keep only** unigrams in $> 1\%$ or $< 99\%$ of documents: 3,672 variables
 - Groups infinite possible posts into “only” $2^{3,672}$ distinct types

Notation

- Document Category

$$D_i = \begin{cases} -2 & \text{extremely negative} \\ -1 & \text{negative} \\ 0 & \text{neutral} \\ 1 & \text{positive} \\ 2 & \text{extremely positive} \\ \text{NA} & \text{no opinion expressed} \\ \text{NB} & \text{not a blog} \end{cases}$$

- Document Category

$$D_i = \begin{cases} -2 & \text{extremely negative} \\ -1 & \text{negative} \\ 0 & \text{neutral} \\ 1 & \text{positive} \\ 2 & \text{extremely positive} \\ \text{NA} & \text{no opinion expressed} \\ \text{NB} & \text{not a blog} \end{cases}$$

- Word Stem Profile:

$$S_i = \begin{cases} S_{i1} = 1 & \text{if "awful" is used, 0 if not} \\ S_{i2} = 1 & \text{if "good" is used, 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "except" is used, 0 if not} \end{cases}$$

Quantities of Interest

Quantities of Interest

- Computer Science: individual document **classifications**

$$D_1, D_2, \dots, D_L$$

Quantities of Interest

- Computer Science: individual document **classifications**

$$D_1, D_2, \dots, D_L$$

- Social Science: **proportions** in each category

$$P(D) = \begin{pmatrix} P(D = -2) \\ P(D = -1) \\ P(D = 0) \\ P(D = 1) \\ P(D = 2) \\ P(D = \text{NA}) \\ P(D = \text{NB}) \end{pmatrix}$$

Issues with Existing Statistical Approaches

Issues with Existing Statistical Approaches

1 Direct Sampling

Issues with Existing Statistical Approaches

- 1 Direct Sampling
 - Biased without a random sample

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

Issues with Existing Statistical Approaches

- ① Direct Sampling
 - Biased without a random sample
 - nonrandomness common due to population drift, data subdivisions, etc.
 - (Classification of population documents not necessary)
- ② Aggregation of model-based individual classifications

Issues with Existing Statistical Approaches

- ① Direct Sampling
 - Biased without a random sample
 - nonrandomness common due to population drift, data subdivisions, etc.
 - (Classification of population documents not necessary)
- ② Aggregation of model-based individual classifications
 - Biased without a random sample

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.
 - $\mathbf{S}$ spans the space of all predictors of D (i.e., all information in the document)

Issues with Existing Statistical Approaches

① Direct Sampling

- Biased without a random sample
- nonrandomness common due to population drift, data subdivisions, etc.
- (Classification of population documents not necessary)

② Aggregation of model-based individual classifications

- Biased without a random sample
- Models $P(D|\mathbf{S})$, but the world works as $P(\mathbf{S}|D)$
- Bias unless
 - $P(D|\mathbf{S})$ encompasses the “true” model.
 - $\mathbf{S}$ spans the space of all predictors of D (i.e., all information in the document)
- Bias even with optimal classification and high % correctly classified

Using Misclassification Rates to Correct Proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions
- (No new assumptions beyond that of the classifier)

Using Misclassification Rates to Correct Proportions

- Use some method to **classify unlabeled documents**
- **Aggregate classifications** to category proportions
- Use labeled set to **estimate misclassification rates** (by cross-validation)
- **Use misclassification rates to correct proportions**
- **Result:** vastly improved estimates of category proportions
- (No new assumptions beyond that of the classifier)
- (still requires random samples, individual classification, etc)

Formalization from Epidemiology

(Levy and Kass, 1970)

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

- Solve:

$$P(D = 1) = \frac{P(\hat{D} = 1) - (1 - \text{spec})}{\text{sens} - (1 - \text{spec})}$$

Formalization from Epidemiology

(Levy and Kass, 1970)

- Accounting identity for 2 categories:

$$P(\hat{D} = 1) = (\text{sens})P(D = 1) + (1 - \text{spec})P(D = 2)$$

- Solve:

$$P(D = 1) = \frac{P(\hat{D} = 1) - (1 - \text{spec})}{\text{sens} - (1 - \text{spec})}$$

- Use this equation to correct $P(\hat{D})$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2008, in press)

Generalizations: J Categories, No Individual Classification

(King and Lu, 2008, in press)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2008, in press)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

- Drop $\hat{D}$ calculation, since $\hat{D} = f(\mathbf{S})$:

$$P(\mathbf{S} = s) = \sum_{j'=1}^J P(\mathbf{S} = s | D = j') P(D = j')$$

Generalizations: J Categories, No Individual Classification

(King and Lu, 2008, in press)

- Accounting identity for J categories

$$P(\hat{D} = j) = \sum_{j'=1}^J P(\hat{D} = j | D = j') P(D = j')$$

- Drop $\hat{D}$ calculation, since $\hat{D} = f(\mathbf{S})$:

$$P(\mathbf{S} = s) = \sum_{j'=1}^J P(\mathbf{S} = s | D = j') P(D = j')$$

- Simplify to an equivalent matrix expression:

$$P(\mathbf{S}) = P(\mathbf{S} | D) P(D)$$

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Document category proportions (quantity of interest)

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Word stem profile proportions (estimate in unlabeled set by tabulation)

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$

Word stem profiles, by category (estimate in *labeled* set by tabulation)

Estimation

The matrix expression again:

$$\underset{2^K \times 1}{P(\mathbf{S})} = \underset{2^K \times J}{P(\mathbf{S}|D)} \underset{J \times 1}{P(D)}$$
$$\implies \mathbf{Y} = \mathbf{X}\beta$$

Alternative symbols (to emphasize the linear equation)

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 & \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

Solve for quantity of interest (with no error term)

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 & \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} & & \end{array}$$

- Technical estimation issues:

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 & \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1

Estimation

The matrix expression again:

$$\begin{array}{ccccc} P(\mathbf{S}) & = & P(\mathbf{S}|D) & P(D) \\ 2^K \times 1 & & 2^K \times J & J \times 1 \\ \implies \mathbf{Y} = \mathbf{X}\beta & \implies & \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \end{array}$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions

Estimation

The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \implies \beta = (X'X)^{-1}X'y$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results

Estimation

The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \quad \implies \quad \beta = (X'X)^{-1}X'y$$

- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results
 - Equivalent to kernel density smoothing of sparse categorical data

Estimation

The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \implies \beta = (X'X)^{-1}X'y$$

- Technical estimation issues:

- 2^K is enormous, far larger than any existing computer
- $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
- Elements of $P(D)$ must be between 0 and 1 and sum to 1

- Solutions

- Use subsets of $\mathbf{S}$; average results
- Equivalent to kernel density smoothing of sparse categorical data
- Use constrained LS to constrain $P(D)$ to simplex

Estimation

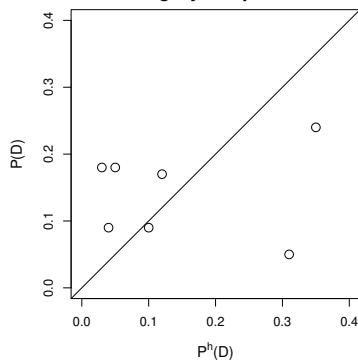
The matrix expression again:

$$\begin{array}{ccc} P(\mathbf{S}) & = & P(\mathbf{S}|D)P(D) \\ 2^K \times 1 & & 2^K \times J \quad J \times 1 \end{array}$$
$$\implies Y = X\beta \quad \implies \quad \beta = (X'X)^{-1}X'y$$

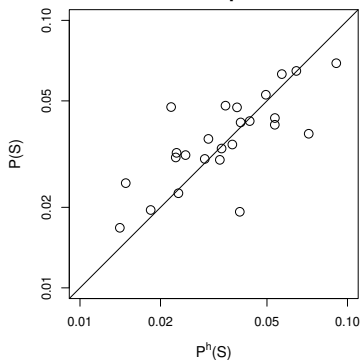
- Technical estimation issues:
 - 2^K is enormous, far larger than any existing computer
 - $P(\mathbf{S})$ and $P(\mathbf{S}|D)$ will be too sparse
 - Elements of $P(D)$ must be between 0 and 1 and sum to 1
- Solutions
 - Use subsets of $\mathbf{S}$; average results
 - Equivalent to kernel density smoothing of sparse categorical data
 - Use constrained LS to constrain $P(D)$ to simplex
- Uncertainty estimates by bootstrapping

A Nonrandom Hand-coded Sample

**Differences in Document
Category Frequencies**

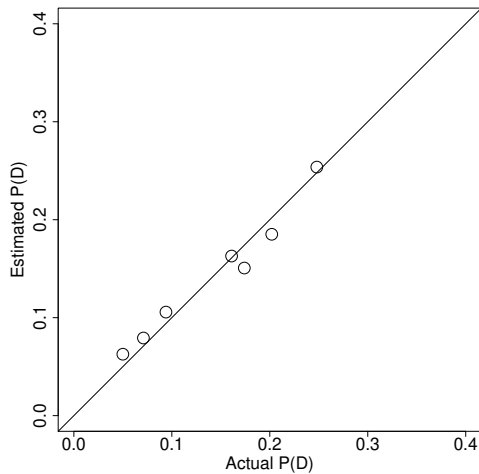


**Differences in Word
Profile Frequencies**

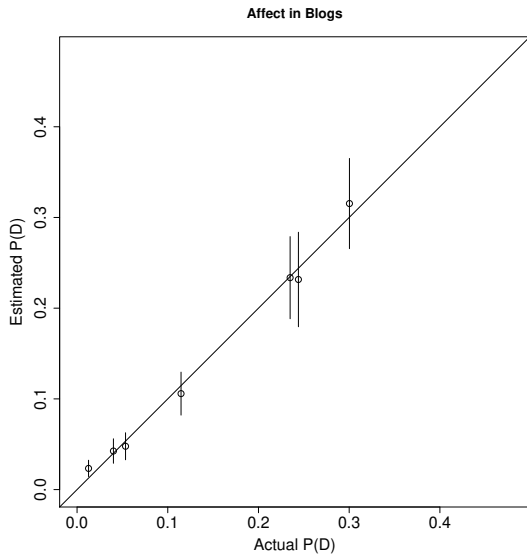


All existing methods would fail with these data.

Accurate Estimates

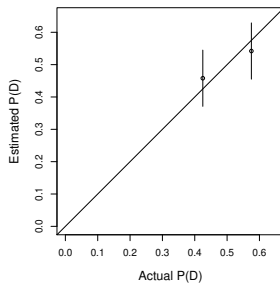


Out of Sample Validation: Blogs

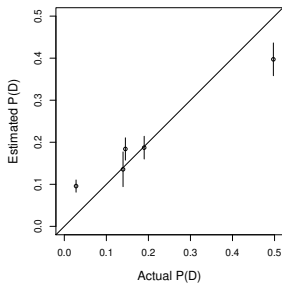


Out of Sample Validation: Other Examples

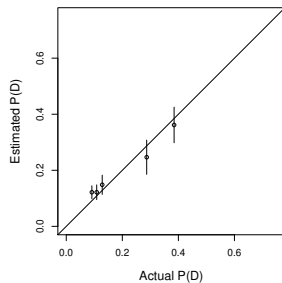
Congressional Speeches



Immigration Editorials



Enron Emails



Verbal Autopsy Methods

Verbal Autopsy Methods

- The Problem

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
 - High quality death registration: only 23/192 countries
- Existing Approaches

Verbal Autopsy Methods

- The Problem
 - Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
 - High quality death registration: only 23/192 countries
- Existing Approaches
 - Ask relatives or caregivers 50-100 symptom questions

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)
- Apply expert algorithms (high reliability, low validity)

- The Problem

- Policymakers need the **cause-specific mortality rate** to set research goals, budgetary priorities, and ameliorative policies
- High quality death registration: only 23/192 countries

- Existing Approaches

- Ask relatives or caregivers 50-100 symptom questions
- Ask physicians to determine cause of death (low intercoder reliability)
- Apply expert algorithms (high reliability, low validity)
- Find deaths with medically certified causes from a local hospital, trace caregivers to their homes, ask the same symptom questions, and statistically classify deaths in population (model-dependent, low accuracy)

An Alternative Approach

An Alternative Approach

- Document-Category, Cause of Death,

$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

An Alternative Approach

- Document ~~Category~~, Cause of ~~D~~Death,

$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

- Word ~~S~~tem Profile, ~~S~~ymptoms:

$$S_i = \begin{cases} S_{i1} = 1 & \text{if "breathing difficulties", 0 if not} \\ S_{i2} = 1 & \text{if "stomach ache", 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "diarrhea", 0 if not} \end{cases}$$

An Alternative Approach

- Document ~~Category~~, Cause of ~~Death~~,

$$D_i = \begin{cases} 1 & \text{if bladder cancer} \\ 2 & \text{if cardiovascular disease} \\ 3 & \text{if transportation accident} \\ \vdots & \vdots \\ J & \text{if infectious respiratory} \end{cases}$$

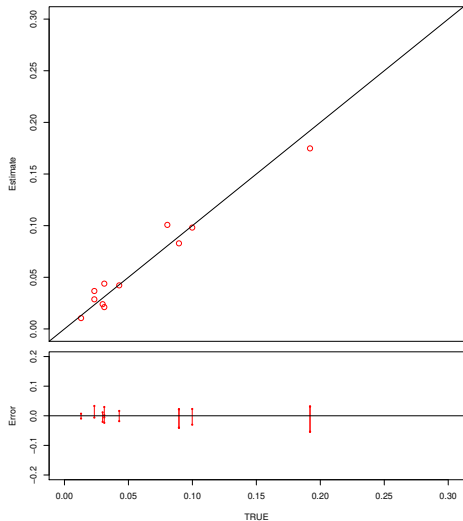
- Word ~~Stem~~ Profile, ~~Symptoms~~:

$$S_i = \begin{cases} S_{i1} = 1 & \text{if "breathing difficulties", 0 if not} \\ S_{i2} = 1 & \text{if "stomach ache", 0 if not} \\ \vdots & \vdots \\ S_{iK} = 1 & \text{if "diarrhea", 0 if not} \end{cases}$$

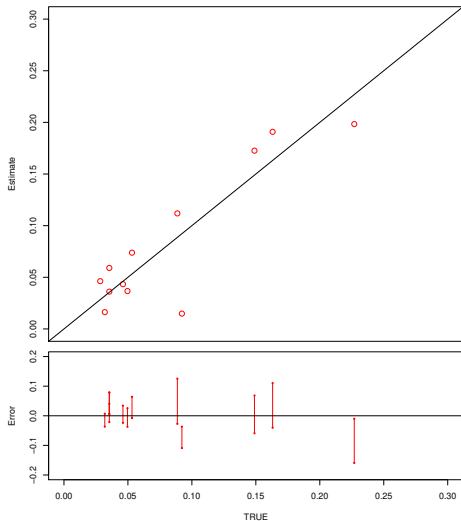
- Apply the ~~same~~ methods

Validation in Tanzania

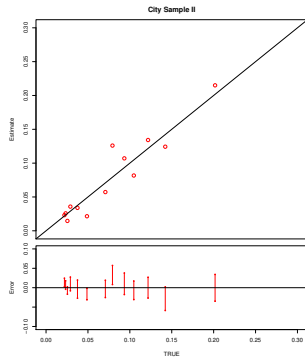
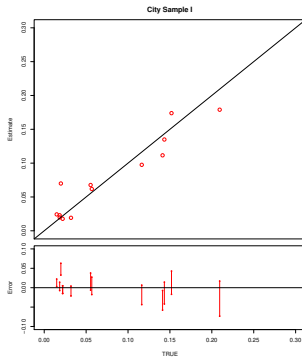
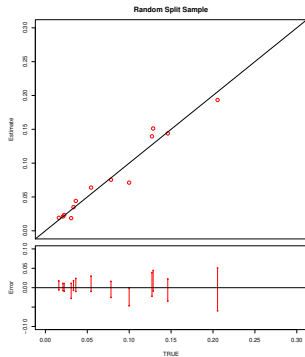
Random Split Sample



Community Sample



Validation in China



Implications for an Individual Classifier

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

$$P(D_\ell | \mathbf{S}_\ell = \mathbf{s}_\ell) = \frac{P(\mathbf{S}_\ell = \mathbf{s}_\ell | D_\ell = j)P(D_\ell = j)}{P(\mathbf{S}_\ell = \mathbf{s}_\ell)}$$

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

$$P(D_\ell | \mathbf{S}_\ell = \mathbf{s}_\ell) = \frac{P(\mathbf{S}_\ell = \mathbf{s}_\ell | D_\ell = j)P(D_\ell = j)}{P(\mathbf{S}_\ell = \mathbf{s}_\ell)}$$

The goal: individual classification

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

$$P(D_\ell | \mathbf{S}_\ell = \mathbf{s}_\ell) = \frac{P(\mathbf{S}_\ell = \mathbf{s}_\ell | D_\ell = j) P(D_\ell = j)}{P(\mathbf{S}_\ell = \mathbf{s}_\ell)}$$

Output from our estimator (described above)

Implications for an Individual Classifier

- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

$$P(D_\ell | \mathbf{S}_\ell = \mathbf{s}_\ell) = \frac{P(\mathbf{S}_\ell = \mathbf{s}_\ell | D_\ell = j)P(D_\ell = j)}{P(\mathbf{S}_\ell = \mathbf{s}_\ell)}$$

Nonparametric estimate from labeled set (an assumption)

Implications for an Individual Classifier

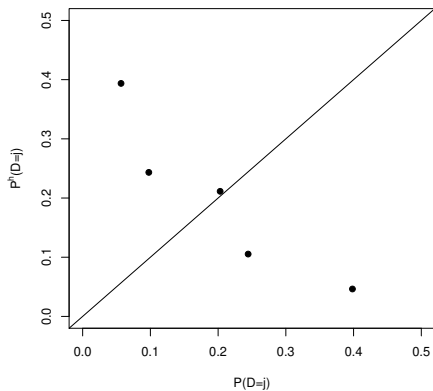
- All existing classifiers assume: $P^h(S, D) = P(S, D)$
- For a different quantity we assume: $P^h(S|D) = P(S|D)$
- How to use this (less restrictive) assumption for classification (Bayes Theorem):

$$P(D_\ell | \mathbf{S}_\ell = \mathbf{s}_\ell) = \frac{P(\mathbf{S}_\ell = \mathbf{s}_\ell | D_\ell = j)P(D_\ell = j)}{P(\mathbf{S}_\ell = \mathbf{s}_\ell)}$$

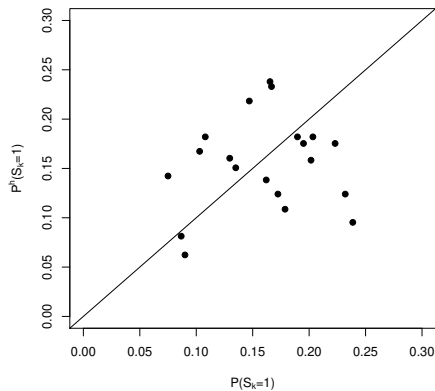
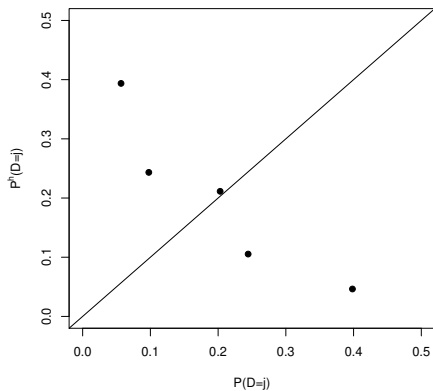
Nonparametric estimate from unlabeled set (no assumption)

Classification with Less Restrictive Assumptions

Classification with Less Restrictive Assumptions

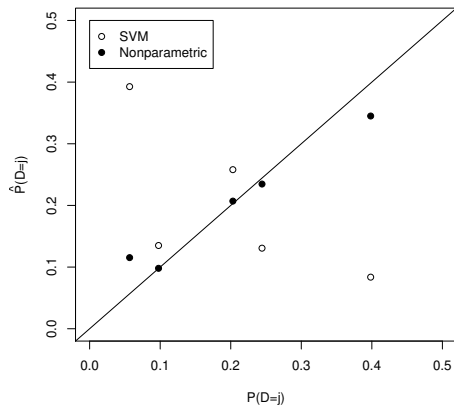


Classification with Less Restrictive Assumptions

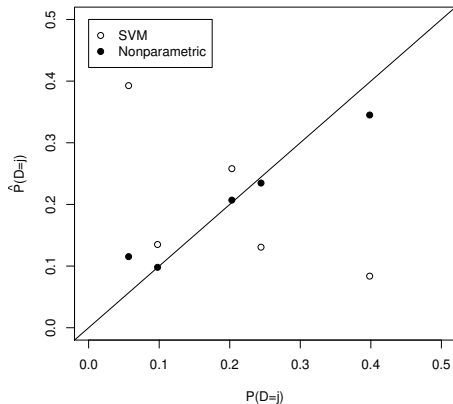


Classification with Less Restrictive Assumptions

Classification with Less Restrictive Assumptions

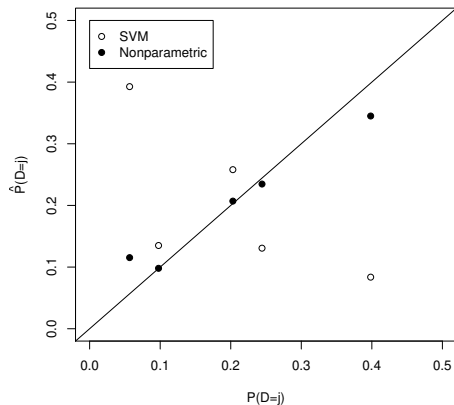


Classification with Less Restrictive Assumptions



Percent correctly classified:

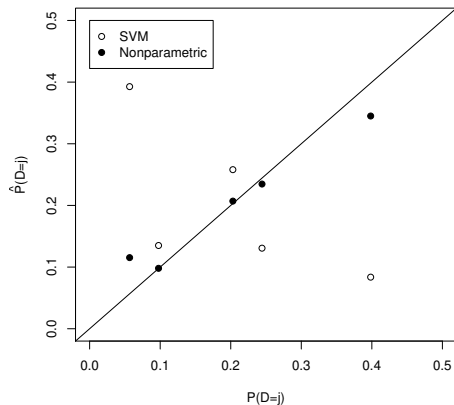
Classification with Less Restrictive Assumptions



Percent correctly classified:

- SVM (best existing classifier): 40.5%

Classification with Less Restrictive Assumptions



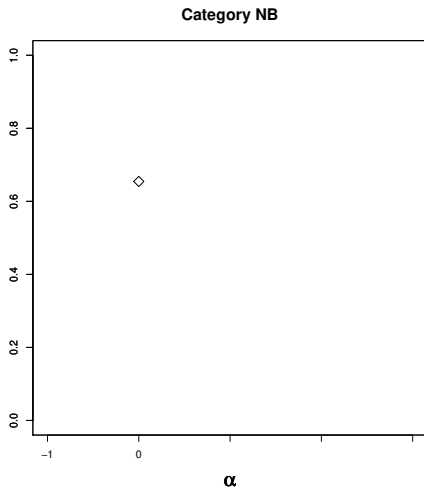
Percent correctly classified:

- SVM (best existing classifier): 40.5%
- Our nonparametric approach: 59.8%

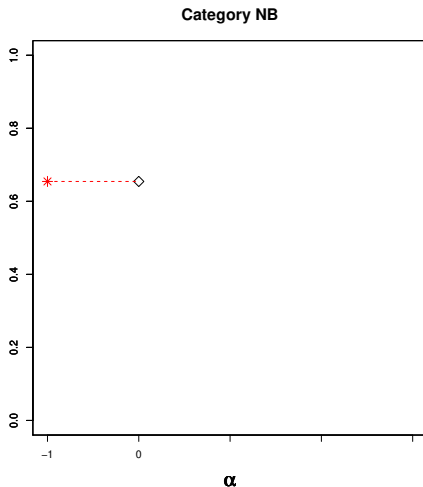
Misclassification Matrix for Blog Posts

	-2	-1	0	1	2	NA	NB	$P(D_1)$
-2	.70	.10	.01	.01	.00	.02	.16	.28
-1	.33	.25	.04	.02	.01	.01	.35	.08
0	.13	.17	.13	.11	.05	.02	.40	.02
1	.07	.06	.08	.20	.25	.01	.34	.03
2	.03	.03	.03	.22	.43	.01	.25	.03
NA	.04	.01	.00	.00	.00	.81	.14	.12
NB	.10	.07	.02	.02	.02	.04	.75	.45

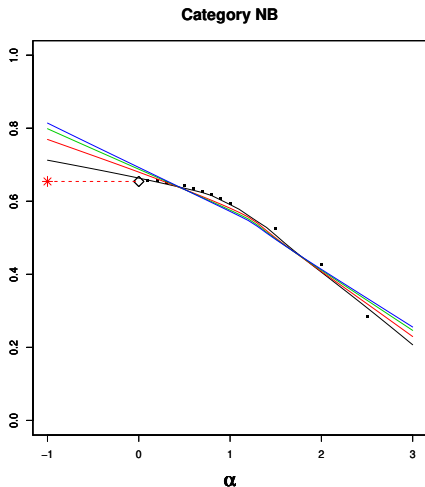
SIMEX Analysis of “Not a Blog” Category



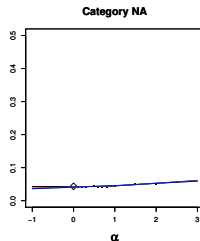
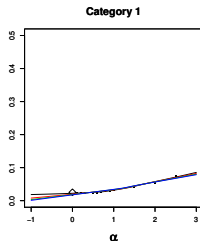
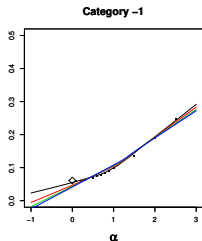
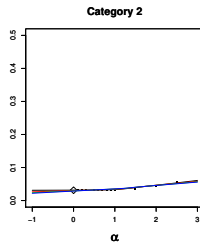
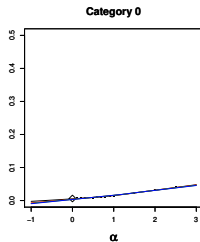
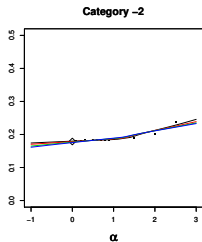
SIMEX Analysis of “Not a Blog” Category



SIMEX Analysis of “Not a Blog” Category



SIMEX Analysis of Other Categories



For more information

<http://GKing.Harvard.edu>

What can go wrong?

What can go wrong?

- We assume $P^h(\mathbf{S}|D) = P(\mathbf{S}|D)$

What can go wrong?

- We assume $P^h(\mathbf{S}|D) = P(\mathbf{S}|D)$
- Must choose word stem subset size (a smoothing parameter)

What can go wrong?

- We assume $P^h(\mathbf{S}|D) = P(\mathbf{S}|D)$
- Must choose word stem subset size (a smoothing parameter)
- Need enough labeled documents in each category (can hand code more if CI's are too large, perhaps via case-control methods)

What can go wrong?

- We assume $P^h(\mathbf{S}|D) = P(\mathbf{S}|D)$
- Must choose word stem subset size (a smoothing parameter)
- Need enough labeled documents in each category (can hand code more if CI's are too large, perhaps via case-control methods)
- Need sufficient information in: documents, categorization scheme, numerical summaries of the documents, and hand-codings

What can go wrong?

- We assume $P^h(\mathbf{S}|D) = P(\mathbf{S}|D)$
- Must choose word stem subset size (a smoothing parameter)
- Need enough labeled documents in each category (can hand code more if CI's are too large, perhaps via case-control methods)
- Need sufficient information in: documents, categorization scheme, numerical summaries of the documents, and hand-codings
- Use additional hand coding to verify assumptions